

UAMM (Uncertainty-Aware Memory Modulation) — Phase 3

P9/P10: max-norm

$$\text{Formula: } \hat{w}_i = \frac{w_i}{\max_j w_j + \varepsilon}$$

Example: $w = [0.30, 0.60, 0.80]$

$$\hat{w} = [0.375, 0.75, \mathbf{1.0}]$$

VS.

P11: softmax $\times m$

$$\text{Formula: } \hat{w}_i = \text{softmax}\left(\frac{1}{\tau_u}\right)_i \times m$$

$$\tau_u = 2.0, \quad m = 3$$

$$\text{Example: logits} = [0, 0, 0] \rightarrow \hat{w} = [1.0, 1.0, 1.0]$$

best modality ALWAYS = 1.0 even if all mod = 1.0; mean=1.0, all independently controlled best: boost > 1.0; worst: suppress < 1.0

P11 UAMM Forward Flow

